

# Divisibility of central binomial coefficients

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| OPG-506 | L1 - Tractable | Ouvert | Number Theory |
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## Énoncé original (anglais, reproduction mot pour mot)

Problem (1) Prove that there exist infinitely many positive integers  $n$  such that  $\gcd(\binom{2n}{n}, 3 \cdot 5 \cdot 7) = 1$ .

Problem (2) Prove that there exists only a finite number of positive integers  $n$  such that  $\gcd(\binom{2n}{n}, 3 \cdot 5 \cdot 7 \cdot 11) = 1$ .

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

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